Microeconomics I

Class 5 — Indirect utility, wealth and price effects

Pavel Kocourek

Universitat de Barcelona · Master in Economics

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Slide 1 Title

Class 5 covers MWG 3.D, 3.G and 2.E: the value of the consumer's problem as a function of prices and wealth, the identity that recovers demand from it, and how demand moves when wealth, its own price or the other price changes. Class 4's best bundle is the input to everything today.

Short quiz

8 minutes · pens down at 10:08

Circle the letter of your answer in the block at the top of the sheet.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

Quiz Start of class, on paper

A short quiz on paper opened the class: circle the letters of your answers in the block on the sheet, write your name on it, and hand in only the strip.

Plan for today

10:00	Short quiz
10:10	Warm-up; rice in Hunan
10:20	Indirect utility
10:41	Roy's identity
10:50	Break
11:00	Wealth effects: normal and inferior goods
11:14	Price effects: Giffen goods, substitutes, complements
11:29	In pairs: from v to demand; summary

Slide 2 Plan for today

The morning in two questions about the best bundle found in class 4: how good it is, which is a number, the indirect utility $v(p, w)$; and how it moves when wealth and prices move. Before the break: v, its properties, and Roy's identity, which turns v back into demand. After the break: wealth effects, own-price effects and cross-price effects, each with a picture.

MWG 3.D, 3.G and 2.E; Muñoz-Garcia chapter 2 is the gentler read.

Practicalities

- Problem set 1: due at the start of class tomorrow, Wed 14 Oct, on paper, one copy per group
- Office hours: Wednesdays 14:00–15:00, Room 445

  up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

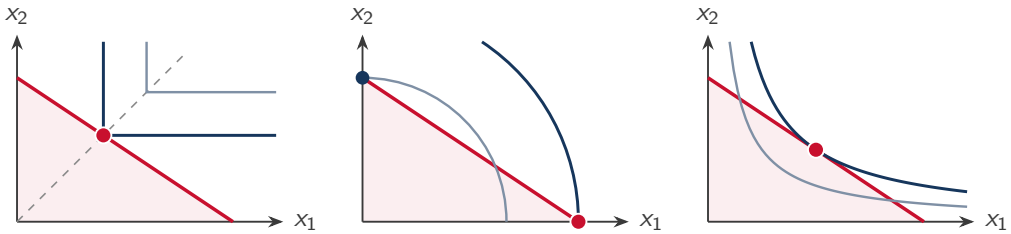
Slide 3 Practicalities

Problem set 1 is due at the start of class tomorrow, Wednesday 14 October, on paper, one copy per group, the answer sheet with all names on top of the working.

Office hours are on Wednesdays, 14:00–15:00, Room 445, up to 20 minutes per student; booking by e-mail gives priority. Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu.

Warm-up: where does each consumer buy?

$p = (2, 3)$, $w = 12$ in every panel



L-shaped, kinks on $x_2 = x_1$ — the kink, not a tangency: $(2.4, 2.4)$

bowed out — a corner, and it jumps: $(0, 4)$ once $p_1 > 3$

$x_1 x_2$ or $\ln x_1 + \ln x_2$ — same bundle, another number: 6 or $\ln 6$

Kinks and corners need no tangency; the formula changes the number, not the bundle.

Rice gets cheaper. Do poor households eat more rice?

Hunan, China, 2006: 650 poor urban households

rice: 64 % of their calories

at random, vouchers: up to a quarter off the price of rice, for five months

More rice, the same, or less?

Jensen & Miller (2008), American Economic Review 98(4): 1553–1577.

Slide 4 Warm-up: where does each consumer buy?

Three consumers face class 4's budget, $p = (2, 3)$, $w = 12$, and the question is where each buys. The point: class 4's tangency rule is a consequence of smooth, convex curves and an interior bundle, not a rule in itself; a kink or a corner can be optimal with no tangency at all.

Panel A: L-shaped curves with kinks on the diagonal are the preference $\min\{x_1, x_2\}$; the curve has no slope at the kink, so the budget slope is irrelevant, and the best bundle is where the budget line crosses the diagonal, $x_1 = x_2 = w/(p_1 + p_2) = 2.4$. Panel B: curves bowed away from the origin are not convex, so along the budget line utility is lowest at the tangency and highest at an end; the optimum is $(6, 0)$ while $p_1 < 3$ and switches to $(0, 4)$ once $p_1 > 3$, so demand jumps there, which is why Roy's identity on slide 13 assumes strictly convex preferences. Panel C: $x_1 x_2$ and $\ln x_1 + \ln x_2$ draw the same map (class 3), so the bundle is $(3, 2)$ either way; only the utility number differs, 6 or $\ln 6$.

That number is what slide 8 names.

Slide 5 Rice gets cheaper. Do poor households eat more rice?

A real price cut for a staple. In 2006, 650 poor urban households in Hunan, China, got about 64 % of their calories from rice; at random, some of them received vouchers worth up to a quarter off the price of rice for five months, the rest were the comparison group. The question put to the room: did the households with cheaper rice eat more rice, the same, or less?

Hold the answer until the next slide. It is a question about how the demand for one good moves with its own price, the second half of today; the trap is to answer from the slope every demand curve is supposed to have.

The study is Jensen and Miller (2008), a field experiment published in the American Economic Review.

Cheaper rice, less rice — for the poor, not the poorest

poor, but not the poorest

at least 20 % of calories from other food

+0.45

% more rice for a 1 % higher price

the poorest

over 80 % of calories from rice

< 0

cheaper rice, more rice

For the poor but not the poorest, demand for rice slopes up: why, after the break.

Jensen & Miller (2008), "Giffen Behavior and Subsistence Consumption", AER 98(4): 1553–1577, Table 3.

Slide 6 Cheaper rice, less rice — for the poor, not the poorest

For households that got at least a fifth of their calories from other food, poor but not the poorest, the estimated price elasticity of rice demand is $+0.45$: rice 1 % dearer, 0.45 % more rice, so cheaper rice meant less rice, and the estimate is significant at the 1 % level. The poorest, living almost on rice alone, did the ordinary thing: cheaper rice, more rice. On the whole sample the estimate is $+0.22$ and not significant: the effect sits exactly where the theory of the second half says it should.

Two cautions. This is one five-month experiment and an estimate with a standard error, not a law. And rice was also inferior for these households: more unearned income, less rice; the link between inferior and upward-sloping demand is slide 22.

The two tiles are Table 3 of Jensen and Miller (2008).

Indirect utility

MWG 3.D, 3.G

Slide 7 Indirect utility

Part 1: the value of the consumer's problem as a function of prices and wealth, and how to get demand back from it. MWG 3.D and 3.G.

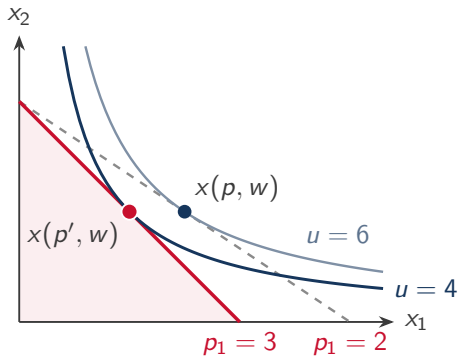
Indirect utility: the level of the best curve the budget reaches

$v(p, w) = u(x^*)$ for any $x^* \in x(p, w)$

the value of the UMP: $\max\{u(x) : x \in \mathcal{B}_{p,w}\}$

$u = x_1 x_2$: $\quad v(p, w) = \dfrac{w}{2p_1} \cdot \dfrac{w}{2p_2} = \dfrac{w^2}{4p_1 p_2}$

at $p = (2, 3)$: $w^2/24$ (class 4, slide 26)



p_1: $2 \to 3$: v falls from 6 to 4.

MWG 3.D, p. 56.

Slide 8 Indirect utility: the level of the best curve the budget reaches

The indirect utility function assigns to each price–wealth pair the utility of the best bundle it affords: a number, not a bundle, and the same number whichever of several best bundles is used. Demand says where the best bundle is; v says how good it is. Both are functions of the three numbers p_1, p_2, w.

For $x_1 x_2$ the formula comes from substituting class 4's demand into u; at $p = (2, 3)$ it is the $w^2/24$ whose slope that class called λ. The picture is the definition read geometrically: a dearer good 1 pivots the budget line inward, the highest reachable indifference curve drops from 6 to 4, and that drop is the fall in v.

What any v must look like, whatever the preference, is the next slide.

$v(p, w)$ is homogeneous, rises with w and never rises with prices

property	why	for $v = \dfrac{w^2}{4p_1 p_2}$
(i) homogeneous of degree 0	the same budget set	$\dfrac{(\alpha w)^2}{4\,\alpha p_1\,\alpha p_2} = v$
(ii) strictly increasing in w	LNS (class 2)	$\dfrac{\partial v}{\partial w} = \dfrac{w}{2p_1 p_2} > 0$
nonincreasing in each p_ℓ	the budget set shrinks	$\dfrac{\partial v}{\partial p_1} = -\dfrac{w^2}{4p_1^2 p_2} < 0$
(iii) quasiconvex in (p, w)	next slide	
(iv) continuous		

MWG Proposition 3.D.3, p. 56: u continuous, $\succsim$ locally non-satiated, $X = \mathbb{R}^2_+$.

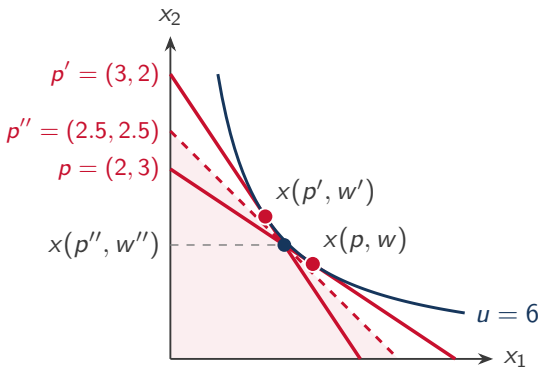
Slide 9 $v(p, w)$ is homogeneous, rises with w and never rises with prices

Proposition 3.D.3 lists what every indirect utility function satisfies, under a continuous u and local non-satiation. Homogeneity of degree zero is inherited from the budget set: scaling prices and wealth together changes nothing, so no assumption on u is needed. Strictly increasing in wealth needs local non-satiation: extra wealth leaves the old best bundle with money to spare, and a strictly better bundle sits close by, still affordable; with a bliss point inside the budget set extra wealth would be worthless.

In prices only nonincreasing, not decreasing: a higher price shrinks the budget set, but a good the consumer does not buy can get dearer without hurting her (class 4's corner solutions). The third column checks each property on the Cobb–Douglas v by hand. Continuity MWG states without proof.

Property (iii), quasiconvexity, is the one that surprises: next slide.

v is quasiconvex: the average of two budgets never beats both



$\{(p, w) : v(p, w) \leq \bar{v}\}$ is convex for every $\bar{v}$

$p = (2,3)$ or $p' = (3,2)$, $w = 12$: $v = 6$ both

their average $(2.5, 2.5)$, $w = 12$: $v = 5.76$

What the average budget affords, one of the two already affords: $v \leq 6$.

MWG Proposition 3.D.3(iii), p. 56; Figure 3.D.5, p. 57, in Wednesday's reading; $u = x_1 x_2$.

In pairs: could this be an indirect utility function?

Test (i) and (ii) of Prop. 3.D.3 (slide 9). Four minutes.

$v(p, w) =$	homogeneous of degree 0	$\uparrow$ in w, $\downarrow$ in p	
$\dfrac{w}{p_1 + p_2}$	✓	✓	$u = \min\{x_1, x_2\}$ (class 4)
$\dfrac{w^2}{p_1 p_2}$	✓	✓	four times slide 8's v
$\dfrac{p_1 + p_2}{w}$	✓	×	more wealth, less utility
$\ln w - \ln p_1 - \ln p_2$	×	✓	changes with the unit of money

Rows 3 and 4 fail: no consumer has them.

Slide 10 — v is quasiconvex: the average of two budgets never beats both

Quasiconvex means: for every level $\bar{v}$, the set of price–wealth pairs at which $v \leq \bar{v}$ is convex. Take two budgets at which $v \leq \bar{v}$ and average them. A bundle affordable at the averaged budget satisfies a weighted sum of the two original budget inequalities, so it satisfies at least one of them: it was already affordable at one of the two budgets, hence gives at most $\bar{v}$. In the picture the dashed averaged line lies inside the union of the two budget sets, and all three lines pass through $(2.4, 2.4)$.

This is quasiconvexity, not quasiconcavity, and nothing about the shape of u was used. Class 3's quasiconcave u had convex upper contour sets in the space of bundles; here the lower contour sets of v, in the space of prices and wealth, are convex. The intuition: extreme price vectors let the consumer specialise in whatever is cheap, so the averaged prices can never beat both originals. The numbers on the slide are the Cobb–Douglas check.

Slide 11 — In pairs: could this be an indirect utility function?

Two tests from Proposition 3.D.3 (slide 9), each a necessary property of any indirect utility function under local non-satiation; a formula that fails either cannot be the v of any such consumer. Homogeneity: replace (p, w) by $(\alpha p, \alpha w)$, prices and wealth together, and see whether α cancels. Direction: the signs of $\partial v / \partial w$ and $\partial v / \partial p_\ell$.

Row 1 is the Leontief consumer's v: she buys the goods one for one, $x_1 = x_2 = w/(p_1 + p_2)$, and $u = \min\{x_1, x_2\}$ is that common quantity (class 4's Exercise 2, row 2). Row 2 is 4 times slide 8's v, the indirect utility of $4x_1 x_2$, the same preference. Row 3 is homogeneous but falls with wealth and rises with prices: no consumer does that, since a larger budget set never lowers the best utility. Row 4 has the right directions but is not homogeneous: $v(\alpha p, \alpha w) = v(p, w) - \ln \alpha$, so in cents instead of euros the consumer would be worse off, the money-illusion mistake of class 4; the price weights must add up to the weight on wealth, as in $\ln w - \frac{1}{2} \ln p_1 - \frac{1}{2} \ln p_2$ on the next slide.

The traps: checking only the directions, and putting α into the prices alone. Quasiconvexity, the third test, is not asked here; rows 1 and 2 pass it because they come from real preferences.

Change the utility function: v and λ change, demand does not

u	$v(p, w)$	$\lambda = \partial v/\partial w$ at $p = (2, 3)$, $w = 12$
$x_1 x_2$	$\dfrac{w^2}{4p_1 p_2}$	$\dfrac{w}{2p_1 p_2} = 1$
$\frac{1}{2}\ln x_1 + \frac{1}{2}\ln x_2$	$\ln w - \frac{1}{2}\ln p_1 - \frac{1}{2}\ln p_2 - \ln 2$	$\dfrac{1}{w} = \dfrac{1}{12}$

Same $\succsim$, same $x(p, w) = (3, 2)$; v becomes $f(v)$, and λ with it.

MWG p. 57, in Wednesday's reading ($\tilde{u} = f(u)$ gives $\tilde{v} = f(v)$; Example 3.D.2 with $\alpha = \frac{1}{2}$); $\lambda = \partial v/\partial w$: pp. 54–55, class 4, slide 26.

Slide 12 Change the utility function: v and λ change, demand does not

Relabel utility by a strictly increasing f and the indirect utility becomes $f(v)$: the best bundle is the same, so its utility is just renamed. The second row is Example 3.D.2 with $\alpha = \frac{1}{2}$, the log form of Cobb–Douglas, where $f(t) = \frac{1}{2}\ln t$.

Class 4's λ, the marginal utility of wealth, now has a home: it is the slope of v in w, by the chain rule and Walras' law. It is rescaled by f' along with v; here $f'(6) = 1/12$ turns 1 into $1/12$. So v and λ are properties of the representation; the bundle is a property of the preference: class 3's ordinal versus cardinal distinction again.

If demand survives every f while v does not, there should be a way to get demand out of v that does not care about f: Roy's identity, next.

Roy's identity: demand from two slopes of v

$$x_\ell(p, w) = -\frac{\partial v(p, w)/\partial p_\ell}{\partial v(p, w)/\partial w}$$

u continuous; $\succsim$ locally non-satiated and strictly convex; v differentiable at $(p, w) \gg 0$

$$v = \frac{w^2}{4p_1 p_2}\colon \quad -\frac{-w^2/(4p_1^2 p_2)}{w/(2p_1 p_2)} = \frac{w}{2p_1}$$

Cobb–Douglas, class 4

Replace u by $f(u)$: both slopes gain the factor f', and the ratio does not change.

MWG Proposition 3.G.4 (Roy's identity), p. 73.

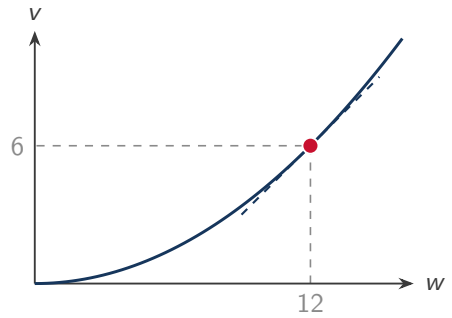
Slide 13 Roy's identity: demand from two slopes of v

Roy's identity recovers each demand from two partial derivatives of v: minus the slope in that good's price, divided by the slope in wealth. The minus sign is there because a higher price lowers v while higher wealth raises it, so the quotient of the raw slopes is negative; a negative demand is the alarm that the sign was dropped.

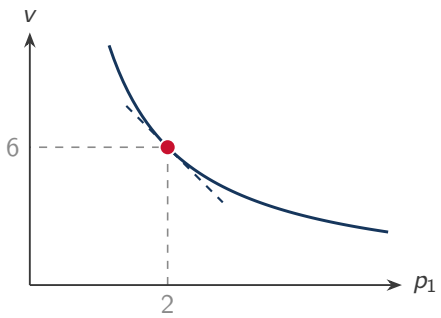
It does not care about the utility representation: replacing u by $f(u)$ multiplies both slopes by the same f', which cancels in the ratio. That is why demand survives relabelling while v and λ do not (slide 12). The assumptions matter: strictly convex preferences make demand a single bundle, and the warm-up's bowed-out consumer, whose demand jumps at $p_1 = 3$, has no Roy's identity there.

Why it is useful: demand by differentiation, with no system of first-order conditions to solve. The next slide says why the formula is true.

A dearer good costs utility in proportion to how much of it she buys



$p = (2, 3)$: slope $\lambda = 1$



$w = 12$, $p_2 = 3$: slope $-\lambda x_1 = -3$

$$\frac{\partial v}{\partial w} = \lambda, \qquad \frac{\partial v}{\partial p_\ell} = -\lambda\, x_\ell(p, w)$$

Here $x_1 = -(-3)/1 = 3$: the bundle $(3, 2)$ again.

MWG p. 74, Proofs 2 and 3 (the envelope theorem); $u = x_1 x_2$, $p = (2, 3)$, $w = 12$.

Slide 14 A dearer good costs utility in proportion to how much of it she buys

The two slopes have meanings of their own. A euro of extra wealth is worth λ in utility (left panel, the picture of class 4's slide 26, slope 1 at $w = 12$). A small rise in p_ℓ makes the same bundle cost x_ℓ euros more per unit of price rise, which is like losing x_ℓ euros of wealth; the consumer does re-optimise, but at an optimum that adjustment has no first-order effect on utility, the envelope theorem, so the loss is λx_ℓ per unit of price (right panel, slope -3 at $p_1 = 2$). Dividing the two slopes gives x_ℓ: Roy's identity.

MWG's second proof is the same argument in formulas: the chain rule gives $\partial v/\partial p_1$ as λ times $\sum_k p_k\, \partial x_k/\partial p_1$, and differentiating the budget identity $p \cdot x(p, w) = w$ in p_1 shows that this sum equals $-x_1$; slide 20 gives that identity a name. The misconception to drop: that the re-optimisation after a price change adds anything of first order.

(optional) A bliss point: does more wealth always raise v?

$u(x) = -(x_1 - 2)^2 - (x_2 - 2)^2$ on $\mathbb{R}^2_+$, $p = (2, 3)$.

(1) $v(p, w)$ for $w \geq 10$.

$v = 0$: $(2, 2)$ costs 10

(2) Which part of Prop. 3.D.3 fails? Why?

(ii): $v(p, 10) = v(p, 12)$; LNS (Def. 3.B.3) fails at $(2, 2)$

(3) $\lambda = \partial v/\partial w$ at $w = 12$?

$\lambda = 0$: the budget does not bind

MWG Proposition 3.D.3, p. 56 (slide 9).

Slide 15 (optional) A bliss point: does more wealth always raise v?

Her indifference curves are circles around $(2, 2)$, the bundle she likes best of all, and at $p = (2, 3)$ it costs 10; from $w = 10$ on she buys it and stops, so v is flat although u is continuous. That is why Prop. 3.D.3(ii) needs local non-satiation and not just continuity: LNS fails at one bundle only, yet once that bundle is affordable it breaks "strictly" for every larger w. For $4/3 \leq w \leq 10$ she buys the point of the budget line closest to $(2, 2)$ and $v = -(10 - w)^2/13$ still rises (below $4/3$: the corner $(w/2, 0)$), so v stays nondecreasing, which slide 9's argument gives without LNS.

At $w = 12$ the multiplier is the slope of a flat v: she spends 10 of her 12 euros, class 4's Kuhn–Tucker case of money left over. The slip is to force a tangency on the line $2x_1 + 3x_2 = 12$: the first-order conditions then give $(30/13, 32/13)$ and $\lambda = -4/13$, and a negative multiplier is the alarm that the budget should not bind.

Break

Back at 11:00

slide 16 of 26

Slide 16 Break

Ten minutes.

Wealth and price effects

MWG 2.E

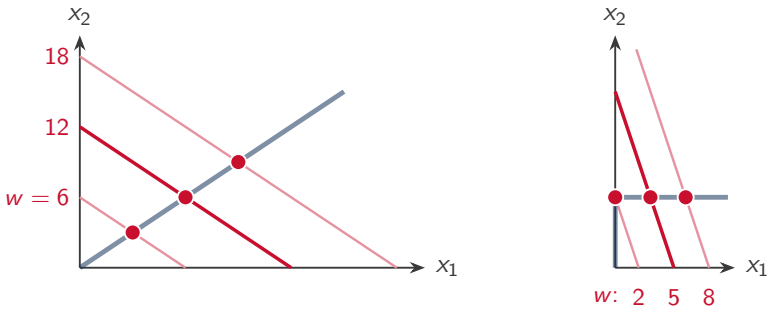
slide 17 of 26

Slide 17 Wealth and price effects

Part 2: the best bundle and its value are known; now wealth moves, then the good's own price, then the other price, and the bundle is watched. MWG 2.E.

More wealth: Cobb–Douglas scales up; quasilinear: all to good 1

Engel function: $w \mapsto x(p, w)$, p fixed; wealth effect $\partial x_\ell / \partial w$



$u = x_1 x_2$, $p = (2, 3)$
$x = (w/4,\ w/6)$: a ray

$u = x_1 + \ln(1 + x_2)$, $p = (3, 1)$
$x_2 = 2$ once $w \geq 2$

Class 4's promise: past $w = 2$, every extra euro goes to good 1.

MWG 2.E, pp. 24–25, Figure 2.E.1 (the wealth expansion path).

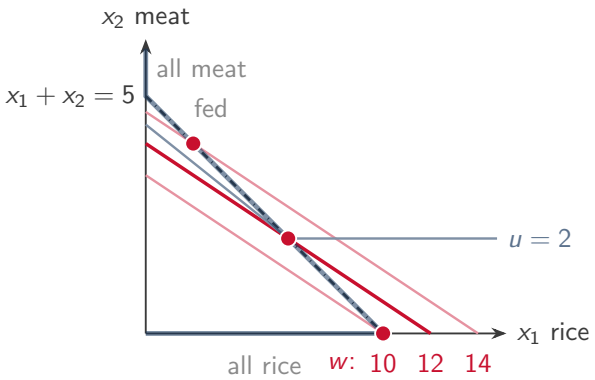
Slide 18 More wealth: Cobb–Douglas scales up; quasilinear: all to good 1

Fix prices and vary wealth: the Engel function; its trace in goods space is the wealth expansion path, MWG Figure 2.E.1 ("income" in other books). Cobb–Douglas spends fixed shares, so the bundle scales along a ray: both wealth effects positive, budget shares constant.

The right panel keeps class 4's promise for $x_1 + \ln(1 + x_2)$ at the prices of its Exercise 3, $p = (3, 1)$. Up to $w = 2$ she buys only good 2; from then on x_2 sits at $p_1/p_2 - 1 = 2$ and every extra euro goes to good 1, so class 4's rows 1 and 2 are two points on this path. Quasilinear utility kills the wealth effect on the nonlinear good once both are bought, which is why it is the workhorse for consumer surplus.

Can a wealth effect be negative? Next slide.

Rice can be inferior: past subsistence, more wealth buys less of it



$p = (2, 3)$

she needs 5 units of food:
$x_1 + x_2 \geq 5$

once fed, only meat counts

normal at (p, w): $\partial x_\ell / \partial w \geq 0$;
inferior: < 0

Rice is normal up to $w = 10$, inferior from 10 to 15.

$u = \min\{x_2,\ x_2 - 4(5 - x_1 - x_2)\}$: meat, minus 4 per unit of missing food. MWG 2.E, p. 25; own example.

Slide 19 Rice can be inferior: past subsistence, more wealth buys less of it

A consumer built for the Hunan question. Rice is the cheap food, meat the dear one; she needs five units of food, each missing unit costs her four utils, and once fed only meat counts. Her indifference curves kink on the food line, so while she can just afford to be fed, the best bundle sits on that line.

Three wealth zones. Below $w = 10$ she cannot reach five units, every euro goes to the cheapest food, and rice is normal. From 10 to 15 she is just fed, and each extra euro swaps one unit of rice for one of meat, which also feeds her, at a cost of $3 - 2 = 1$ euro: rice falls one for one with wealth, inferior, while meat is normal. Above 15 she is fed on meat alone and buys no rice.

Normal and inferior are properties at a given (p, w), not of a good forever; MWG notes that inferiority is common for narrow goods, through substitution toward higher quality as wealth rises. MWG counts a zero wealth effect as normal.

The budget ties the effects together: not every good can be inferior

$$\sum_k p_k \frac{\partial x_\ell}{\partial p_k} + w \frac{\partial x_\ell}{\partial w} = 0 \qquad \text{homogeneity: Prop. 2.E.1}$$

$$\sum_\ell p_\ell \frac{\partial x_\ell}{\partial w} = 1 \qquad \text{Walras in } w\text{: Engel aggregation, Prop. 2.E.3}$$

$$\sum_\ell p_\ell \frac{\partial x_\ell}{\partial p_k} + x_k = 0 \qquad \text{Walras in } p_k\text{: Cournot aggregation, Prop. 2.E.2}$$

Slide 19 at $w = 12$: $2 \cdot (-1) + 3 \cdot (+1) = 1$. An extra euro is spent somewhere.

MWG 2.E, pp. 27–28.

Slide 20 The budget ties the effects together: not every good can be inferior

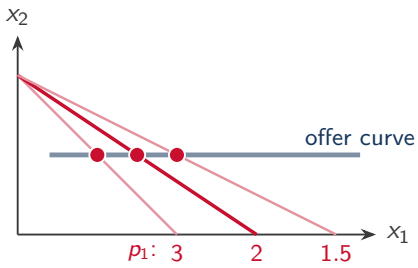
Three identities that every Walrasian demand satisfies, each the derivative of something known. Differentiate homogeneity $x(\alpha p, \alpha w) = x(p, w)$ in α: line 1. Differentiate Walras' law $p \cdot x(p, w) = w$ in w: line 2, Engel aggregation; in p_k: line 3, Cournot aggregation, which was the step inside Roy's proof on slide 14.

Line 2 says an extra euro is spent somewhere: the price-weighted wealth effects add to one, so with two goods at most one can be inferior. The slide's check is slide 19's consumer at $w = 12$, where rice loses one unit and meat gains one. MWG also writes these identities with elasticities (Exercise 2.E.2).

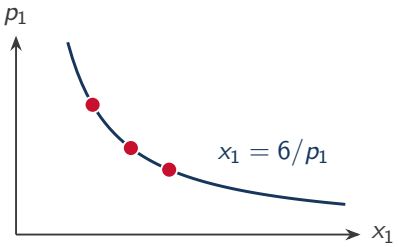
The same consumer checks the other two lines. Her rice demand near $p = (2, 3)$, $w = 12$ is $x_1 = (5p_2 - w)/(p_2 - p_1)$, with $\partial x_1/\partial p_1 = 3$, $\partial x_1/\partial p_2 = 2$ and $\partial x_1/\partial w = -1$, so line 1 reads $2 \cdot 3 + 3 \cdot 2 + 12 \cdot (-1) = 0$; with $\partial x_2/\partial p_1 = -3$, line 3 for $k = 1$ reads $2 \cdot 3 + 3 \cdot (-3) + 3 = 0$.

Cheaper good 1: Cobb–Douglas buys more of it, as much of good 2

usual: $\partial x_\ell/\partial p_\ell \leq 0$; Giffen at (p, w): $\partial x_\ell/\partial p_\ell > 0$



goods: the offer curve



prices: the demand curve

$x_1 = 6/p_1$ rises as p_1 falls; x_2 stays at 2: the offer curve is flat.

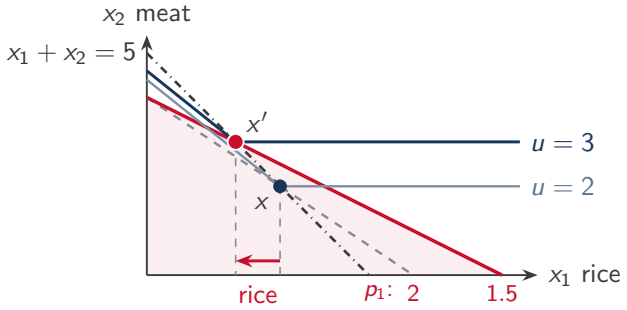
MWG 2.E, pp. 25–26, Figures 2.E.2–2.E.3; $u = x_1 x_2$, $w = 12$, $p_2 = 3$.

Slide 21 Cheaper good 1: Cobb–Douglas buys more of it, as much of good 2

The own-price effect, read two ways. Left, in goods space: as p_1 falls the budget line pivots outward around $(0, 4)$, and the best bundles trace the offer curve, MWG Figure 2.E.3. Cobb–Douglas spends half of wealth on good 1 whatever its price, so $x_1 = 6/p_1$ rises while x_2 stays at 2: class 4's "nothing happens to x_2" is the flat offer curve.

Right, the same information as a demand curve with the price on the vertical axis, the economists' habit (MWG p. 25). The law of demand says it slopes down; MWG's definition allows the opposite, a Giffen good at (p, w), "not an economic impossibility". Whether that can happen is the next slide, back with slide 19's consumer.

Cheaper rice, less rice: the subsistence consumer is Giffen



p_1: $2 \to 1.5$

rice $3 \to 2$, meat $2 \to 3$

on the food line: $x_1 = 3/(3 - p_1)$ rises with p_1

Giffen needs inferior (slide 19)

Cheaper rice frees money for meat, which also feeds her.

Jensen & Miller (2008), AER 98(4): their subsistence model; own numbers, $w = 12$, $p_2 = 3$.

Dearer good 2: x_1 falls, stays or rises, depending on σ

gross substitutes: $\partial x_1/\partial p_2 > 0$; gross complements: < 0

CES: $u = \left(\frac{1}{2}x_1^\rho + \frac{1}{2}x_2^\rho\right)^{1/\rho}$, $\sigma = 1/(1-\rho)$; $\sigma = 2$: same $\gtrless$ as $\sqrt{x_1} + \sqrt{x_2}$

σ	u	$x_1(p, w)$	p_2: $3 \to 4$ $p_1 = 2$, $w = 12$
0	$\min\{x_1, x_2\}$	$\dfrac{w}{p_1 + p_2}$	$2.4 \to 2$: complements
1	$x_1 x_2$	$\dfrac{w}{2p_1}$	$3 \to 3$: neither
2	$\sqrt{x_1} + \sqrt{x_2}$	$\dfrac{w\, p_2}{p_1(p_1 + p_2)}$	$3.6 \to 4$: substitutes

CES: complements for $\sigma < 1$, neither at $\sigma = 1$, substitutes for $\sigma > 1$.

MWG 3.G, p. 70 ("gross" = Walrasian demand; the Hicksian version is Wednesday). CES: class 3, slide 26.

Slide 22 Cheaper rice, less rice: the subsistence consumer is Giffen

Slide 19's consumer, $w = 12$, $p_2 = 3$; rice falls from 2 to 1.5, a quarter off, the largest Hunan voucher. She stays exactly fed, now at $(2, 3)$, rice $3 \to 2$ and meat $2 \to 3$: being fed costs less, the money saved buys meat, and meat also feeds her, so she needs less rice. On the food line her rice demand is $3/(3 - p_1)$, rising in p_1 while she can be fed: Giffen.

Jensen and Miller's model has the same three zones; their data split the households at 80 % of calories from rice: below $w = 10$ she is like their poorest, all rice, ordinary demand; between 10 and 15 like the poor but not the poorest: Giffen. Their top zone, where rice demand slopes down again (ours buys no rice at all), they could not separate in a sample of the very poor. The mechanism needs rice to be inferior (slide 19): a cheaper staple makes her effectively richer, and extra wealth buys less of it; a Giffen good must be inferior, not the converse, and the Slutsky equation next Tuesday shows why. MWG's potato story (page 26) is the same mechanism.

The numbers here are illustrative; the 0.45 is an estimate for one group in one five-month experiment.

Slide 23 Dearer good 2: x_1 falls, stays or rises, depending on σ

The cross-price effect with wealth fixed, which MWG calls "gross": good 1 is a gross substitute for good 2 if a dearer good 2 raises x_1, a gross complement if it lowers it. The three rows are class 4's CES family at the same points 2.4, 3, 3.6 as on its slide 29. Leontief buys the goods as a pair, so a dearer pair means less of both: complements. Cobb–Douglas spends $w/2$ on good 1 whatever p_2: neither. At $\sigma = 2$ she leans away from the dearer good: substitutes. In general the sign of $\partial x_1/\partial p_2$ is the sign of $\sigma - 1$.

Gross relations need not run both ways: for slide 22's consumer dearer meat means more rice, but dearer rice means less meat, so the pair is substitutes in one direction and complements in the other. MWG's own definition of substitutes uses Hicksian demand, and that net version comes with Hicksian demand on Wednesday 14 October; the net relation is symmetric, which the Slutsky matrix shows next Tuesday. MWG writes weak inequalities; with strict ones a zero effect is "neither".

In pairs: Roy's identity, then the three effects

Find the demand by Roy's identity. Wealth effect? Own price: usual or Giffen? The other price rises: substitutes or complements? Seven minutes.

$v(p, w)$	demand	wealth	own price	other price ↑
$\frac{w^3}{p_1 p_2^2}$, find x_1	$x_1 = w/(3p_1)$	normal	usual	neither
$\frac{w+p_2}{p_1} - 1 + \ln\frac{p_1}{p_2}$, find x_2 class 4's $x_1 + \ln(1+x_2)$, $p_1 > p_2$	$x_2 = p_1/p_2 - 1$	zero: normal in MWG	usual	substitutes
$\frac{w-5p_1}{p_2-p_1}$, find x_1 slide 22's consumer, near $p = (2,3)$, $w = 12$	$x_1 = \frac{5p_2 - w}{p_2 - p_1}$	inferior	Giffen	substitutes

Row 3 is slide 22's rice: inferior and Giffen, from one quotient.

Slide 24 In pairs: Roy's identity, then the three effects

The recipe: Roy's identity first, minus the price slope over the wealth slope, then three signs: $\partial x/\partial w$ for the wealth effect, the own-price slope for usual or Giffen, the other-price slope for substitutes or complements.

Row 1 is a Cobb–Douglas consumer with a third of wealth on good 1 ($u = x_1 x_2^2$ up to a factor): normal, usual, and p_2 does not appear in x_1 although it appears in v, so neither; the trap is to read a cross effect off v instead of x. Row 2 is class 4's $x_1 + \ln(1+x_2)$ by another road: $x_2 = p_1/p_2 - 1$ has a zero wealth effect, which MWG still counts as normal, falls in its own price, and rises in p_1: substitutes. Row 3 is slide 22's rice by the quotient rule: $x_1 = (5p_2 - w)/(p_2 - p_1)$, which equals 3 at $p = (2,3)$, $w = 12$, falls with w (inferior), rises with p_1 (Giffen) and rises with p_2 (substitutes), all from one formula.

The slips to watch: dropping the minus sign (a negative demand is the alarm), dividing by the price slope instead of the wealth slope, calling a zero wealth effect inferior, and calling any inferior good Giffen; Giffen implies inferior, not the converse.

(optional) Good 1 is Giffen: what happens to x_2 when p_1 rises?

A consumer with two goods; Walras' law holds; $\partial x_1/\partial p_1 > 0$.

(1) Show that $\partial x_2/\partial p_1 < 0$. — $p_2\,\partial x_2/\partial p_1 = -x_1 - p_1\,\partial x_1/\partial p_1 < 0$

Hint: slide 20, line 3, with $k = 1$.

(2) Explain (1) with her spending. — p_1 and x_1 rise, so $p_1 x_1$ rises and $p_2 x_2$ falls

(3) Can $\partial x_1/\partial p_2 > 0$ hold too? — Yes: slide 22's rice at $p_1 = 2$: $\partial x_1/\partial p_2 = 2$, $\partial x_2/\partial p_1 = -3$

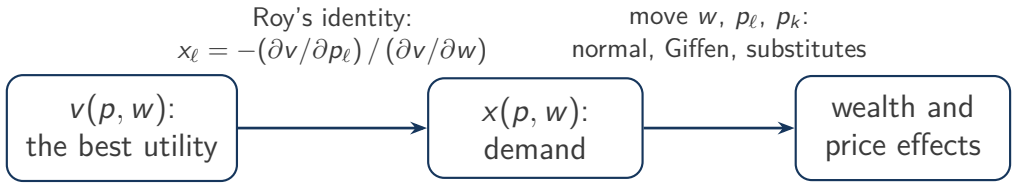
MWG 2.E, pp. 27–28 (Proposition 2.E.2, Cournot aggregation).

Slide 25 (optional) Good 1 is Giffen: what happens to x_2 when p_1 rises?

Line 3 of slide 20 with $k = 1$ is Walras' law differentiated in p_1, so it holds for every consumer who spends all her wealth; move the two good-1 terms to the right and both are negative once $\partial x_1/\partial p_1 > 0$, which is why Walras' law alone fixes the sign and "Giffen implies inferior" is never needed. The term that is easy to drop is x_1, the extra cost of the units she already buys; in words, the identity says that spending on a good whose price and quantity both rise must go up, leaving less for the other good when wealth is fixed. Slide 22's rice at $p = (2,3)$, $w = 12$ gives the size: p_1 from 2 to 2.1 moves spending on rice from 6 to 7 euros and meat from 2 to $1\frac{2}{3}$ units, the $\partial x_2/\partial p_1 = -3$ used in slide 20's commentary.

Part (3) is slide 23's asymmetry on the same consumer: dearer meat means more rice, $\partial x_1/\partial p_2 = 2$, while dearer rice means less meat, so rice is a gross substitute for meat but meat a gross complement for rice, and "good 1 is a substitute for good 2" does not carry over to good 2. The net relation, read off Hicksian demand, is symmetric: the Slutsky matrix on Tuesday 20 October.

Today in one line



Wednesday: expenditure minimisation — Hicksian demand, Shephard's lemma, duality.
Problem set 1: due at the start of class tomorrow, on paper, one copy per group.
Problem set 2 is handed out tomorrow; due in class Wednesday 21 October.

Read MWG 3.E–3.G (or Muñoz-Garcia ch. 2) for Wednesday 14 October.

Slide 26 Today in one line

In one line: $v(p, w)$ is the value of the consumer's problem, homogeneous of degree zero, increasing in wealth, nonincreasing in prices, quasiconvex, and it changes with the utility representation; Roy's identity gets demand back from it with two derivatives, and demand does not change with the representation. Demand moves with wealth (normal or inferior, not everything inferior), with its own price (usually down, Giffen only if inferior) and with the other price (gross substitutes or complements). Hunan's rice did all of this.

Wednesday 14 October: the dual problem, the cheapest way to reach a utility level: Hicksian demand, the expenditure function, Shephard's lemma, duality, and the true cost-of-living index. Reading: MWG 3.E–3.G, or Muñoz-Garcia chapter 2.

Problem set 1 is due at the start of class tomorrow, on paper, one copy per group; problem set 2 is handed out tomorrow and is due in class on Wednesday 21 October.